INTRODUCTION

Credit analysis plays a critical role in fixed-income markets. Proper evaluation and pricing of credit risk facilitates the efficient allocation of capital. This is a dynamic process as credit risk components are continuously re-evaluated and fixed-income instruments repriced according to market conditions. This learning module covers the basic principles of credit analysis. First, we introduce the concepts of credit risk and expected loss and interpret what credit ratings mean. We compare bond issuer creditworthiness within a given industry as well as across industries, and we explore how financial markets price credit risk. This lesson focuses primarily on the analysis of credit risk, while subsequent lessons discuss credit analysis of sovereign and non-sovereign government issuers as well as corporate debt

Learning Module Overview

- Credit risk is the risk of economic loss resulting from borrower failure to make full and timely payments of interest and principal. The key components of credit risk are the probability of default and the loss given default, and their product is expected loss.
- Chief sources of credit risk include adverse macroeconomic conditions, a financing mismatch between resources and obligations, and issuer-specific factors in corporate and sovereign debt markets.
- Nearly every bond issue in developed debt markets carries credit ratings classifying creditworthiness. Credit ratings enable comparisons of the credit risk of debt issues and issuers within and across industries.
- Bonds or issuers with an investment-grade (IG) credit rating pose the lowest risk of default and are rated Baa3 by Moody's and BBB– or higher by S&P and Fitch. In contrast, non-investment-grade or high-yield (HY) bonds or issuers are rated BB+ or lower by S&P/Fitch and Ba1 or less by Moody's and represent substantial to very high credit risk.
- Pitfalls of relying solely on credit ratings in making investment decisions include that rating agency decisions may lag market pricing of credit risk, overlook key financial risks, and/or involve miscalculations or unforeseen changes not fully captured in a rating agency's forward-looking analysis.
- The premium, or yield spread, at which corporate bonds trade relative to default risk-free assets widens when credit risk rises and narrows if credit risk falls.
- Credit spread changes affect holding period returns via two primary factors: a) the basis point spread change and b) the sensitivity of price to yield as reflected by end-of-period modified duration and convexity. Spread narrowing increases holding period returns.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. Credit risk is:

C. experienced in several ways by lenders.

Hide Solution

Solution:

The correct answer is C. Lenders are impacted by credit risk in several ways, including failure to receive principal and interest payments in a timely manner, the inability to sell collateral at a market price sufficient to meet an issuer's obligations in the case of secured debt, or the potential incurrence of legal or other costs to collect debt. A is incorrect because while a borrower default initiates a realized loss, expected loss exists prior to the occurrence of any actual loss. B is incorrect because it refers to the credit spread, which compensates investors for assuming credit risk.

2. A EUR500,000 loan has the following characteristics:

- Probability of default 5%
- Collateral EUR100,000
- Recovery rate 90%
- Expected exposure EUR400,000

The expected loss for this loan in event of default is:

A. EUR1,500
B. EUR2,000
C. EUR20,000

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Solution:

The correct answer is B. We solve for expected loss (EL) as follows:

EL = POD × (EE – Collateral) × (1 – RR).

Since probability of default (POD) is 5%, expected exposure (EE) is EUR400,000, collateral is EUR100,000, and the recovery rate (RR) is 90%:

EL = EUR1,500 = 0.05 × (400,000 – 100,000) × (1 – 0.9).

A is incorrect as it fails to reduce the expected exposure by the collateral, while C is incorrect as it simply multiplies EE and POD.

3. Credit ratings are:

A. developed on behalf of investors.
B. a symbol-based measure of the potential default risk of a bond issue or issuer.
C. measures of credit risk that are used to determine bond market pricing.

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Solution:

The correct answer is B: Credit ratings are a symbol-based measure of the potential default risk of a bond issue or issuer. A is incorrect because credit ratings are developed on behalf of the issuer. C is incorrect because credit ratings are not used to determine bond prices.

4. Which of the following statements best describes risks that are difficult to capture in credit ratings?

A. Environmental risks are captured by ESG ratings rather than credit ratings.
B. Debt-financed acquisitions are usually captured in credit ratings.
C. Split ratings demonstrate that credit rating agencies may view complex risks very differently.

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The correct answer is C. A is incorrect as environmental risks may affect credit ratings. B is incorrect since debt-financed acquisitions are difficult to anticipate.

5. *Determine the correct answers to fill in the blanks:* Spreads are _____ at or near the top of the credit cycle, when market participants perceive credit risk to be at its lowest; they are _____ at or near the bottom of the credit cycle, when financial markets believe credit risk has reached its peak.

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Solution:

Spreads are **narrowest (or lowest)** at or near the top of the credit cycle, when market participants perceive credit risk to be at its lowest; they are **widest (or highest)** at or near the bottom of the credit cycle, when financial markets believe credit risk has reached its peak.

6. A portfolio manager assessing a downside case believes that HY bond spreads will rise 100 bps in a recession. If an observed HY bond has modified duration of 4.5 and reported convexity of 0.23, the expected change in the HY bond's price under this scenario would be *closest to*:

A. –4.5%.
B. –4.385%.
C. –4.615%.

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Solution:

The correct answer is B. Bond price changes based upon modified duration, convexity, and the spread change are calculated as follows:

$$\%\Delta PV^{Full} = -(\text{AnnModDur} \times \Delta\text{Spread}) + \tfrac{1}{2}\text{AnnConvexity} \times (\Delta\text{Spread})^2.$$

With AnnModDur = 4.5 and reported convexity of 0.23, rescaling convexity to 23 and substituting values to solve for spread results in

$$\Delta PV^{Full} = -(4.5 \times 1.00\%) + \tfrac{1}{2}(23) \times (1.00\%)^2$$

$$\Delta PV^{Full} = -4.385\%.$$

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HG For Q2 of the self assesment, there is a mistake in the answer which should be A

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AJ In Q2, they don´t ask about the expected loss of the loan; they ask about the expected loss IN THE EVENT OF DEFAULT. Should we not then assume probability of default is 100%? In that case, expected loss = expected exposure x (1 - recovery rate) = 40.000. Someone had the same doubt? Or am I overthinking it?

Created 3 months ago by Alvaro Jose Moreno Morillo 2 replies | Last Activity: 2 months ago

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PK Q-6 GIVEN Convexity is 0.23 but its need to be 23 to match ans.

Created 4 months ago by pritesh kumar kesarbhai chaudhary

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